

PROJECT CANOPY

DynamicBoost + DCAC 2022
12kW boost + DC-AC converter
High-density power stage

Range-Ext E-Scooter 2021
Range-extended e-scooter drive
3kW, validated vs Ansys Twins

ISG Controller 2020
Automotive ISG, CAN diagnostics
Passed 100A at 12krpm

Two-segment PMSM 2018
Fault-tolerant six-phase PMSM
Fault-tolerant six-phase control

Generator 2016
Portable low-THD engine generator
THDv below 0.8%

Refrigerator 2023
Sensorless compressor drive
High efficiency at low speed

E-Scooter 2022
Mass-production 6kW scooter drive
Cost-down on a lower-end MCU

Embedded System 2020
Embedded base layer from scratch
Reusable linker scripts + IVT

PMASynRM 2019
Non-rare-earth PM-assisted SynRM
0v control cuts torque ripple

PV Inverter 2017
Solar PV grid-tie inverter, MPPT
99.5% MPPT efficiency

SKILL ROOTS

● Embedded Firmware

- **Languages:** ▸ Assembly ▸ C
- **MCU Platforms:** ▸ TI-DSP
- Renesas ▸ Microchip ▸ ST ▸ NXP
- **Embedded Functions:**
- Embedded System ▸ Bootloader
- **Communication:** ▸ SPI ▸ UART
- **CAN:** ▸ ISO15762 ▸ ISO14230

● Modelling & Digital Twin

- **System Modelling:** ▸ Ansys Twins
- MATLAB
- **Circuit Modelling:** ▸ Ansys Q3D
- SIMetrix ▸ PSpice ▸ PSIM

● AI Engineering

- **Analysis Automation:**
- Dynamic Analysis (MATLAB)
- Python & GUI

● Control & Power Conversion

- Engine Control
- **DC Bus:** ▸ PAM ▸ CV/CC
- **Grid / AC Load:** ▸ MPPT ▸ PFC
- Droop Control
- **Motor Control:** ▸ Capless
- Speed Control ▸ Position Control
- MTPA ▸ Flux Weakening
- 0v Control ▸ FOC ▸ V/F ▸ I/F
- **Position Sensing:**
- Sensorless / Back-EMF
- Hall Sensor ▸ Encoder ▸ Resolver
- **Modulation:** ▸ SPWM ▸ SVPWM
- DPWM1 ▸ DPWM Min
- Six-Step / One-Pulse

● Engineering Process & Standards

- **Standards:** ▸ ISO 26262 ▸ ASPICE
- Static Analysis (Polyspace)

● Hardware & Test Systems

- **Motor Types:** ▸ PMSM ▸ IM
- Six-Phase PMSM ▸ PMASynRM
- SynRM
- **Power Components:** ▸ IGBT
- MOSFET ▸ GaN ▸ SiC
- **Hardware Layout:** ▸ Peripheral
- Power Stage
- **Dynamometer:** ▸ **Active Load:**
- Yaskawa ▸ YL ▸ Delta ▸ Siemens
- Passive Load (EM Brake)
- **Shaft Alignment:** ▸ Dial Indicator
- Laser Alignment
- **Measurement:**
- Motor Characteristics (R, L, Ke, Kt, J)
- System / Circuit (Bode, FFT)